

Grade 9 Ontario Mathematics Test

Duration: 60 minutes Total Marks: 50

Name: _____ Date: _____

Part A: Number Sense & Algebra (10 marks)

1. Simplify the expression:

$$3(2x - 4) + 5(x + 1) \quad [2 \text{ marks}]$$

2. Solve for x:

$$2x + 5 = 3x - 4 \quad [2 \text{ marks}]$$

3. Expand and simplify:

$$(x - 3)(x + 2) \quad [2 \text{ marks}]$$

4. Factor:

$$x^2 + 5x + 6 \quad [2 \text{ marks}]$$

5. Evaluate:

$$(3^2 + 4^2) / (7 - 2) \quad [2 \text{ marks}]$$

Part B: Linear Relations (10 marks)

6. A line passes through the points (2, 5) and (6, 13).

a) Find the slope.

b) Write the equation in slope-intercept form. [4 marks]

7. Determine whether the following table represents a linear relationship:

x | y

1 | 3

2 | 5

3 | 7

4 | 10 [2 marks]

8. Graph the equation $y = -2x + 4$ on the provided grid. [4 marks]

Part C: Geometry & Measurement (10 marks)

9. A triangle has side lengths 6 cm, 8 cm, and 10 cm.

a) Is this a right triangle? Show your work. [2 marks]

10. Find the area and perimeter of a rectangle with length 12 cm and width 5 cm. [2 marks]

11. A cylinder has a radius of 3 cm and height of 10 cm.

Find the volume. (Use $\pi = 3.14$) [3 marks]

12. Convert 2.5 hours into minutes and seconds. [1 mark]

13. Convert 5.2 km into meters. [1 mark]

Part D: Consumer Math (20 marks)

14. A TV is originally priced at \$850. It is on sale for 20% off.

a) What is the discount?

b) What is the sale price? [4 marks]

15. After tax (13% HST), what is the total cost of the TV from question 14? [2 marks]

16. You earn \$15.75 per hour. You work 38 hours in a week.

a) What is your gross weekly income?

b) If 25% is deducted for taxes and benefits, what is your net pay? [4 marks]

17. You buy a \$1,200 laptop using a credit card. The interest rate is 19.99% annually,
but you only make the minimum monthly payment of \$30.

Estimate how long it will take to pay off and how much interest you might pay over time. (Explain your reasoning without a calculator.) [4 marks]

18. Budgeting: Your monthly income is \$1,600. You spend:

- Rent: \$600

- Food: \$250

- Transit: \$120

- Entertainment: \$130

- Savings: \$150

a) How much is left after expenses?

b) What percentage of your income is spent on rent? [6 marks]